FraudLens

Financial Transaction & Fraud Investigation Management System

Project Focus & Submission Scope

Review 1 Evaluation Report: Relational Domain Modeling, Third Normal Form (3NF) Architecture, Oracle SQL DDL Schema with Named Constraints & Sequences, Controlled DML Operations, Single-Row Formatting, Set Operators, Aggregates, and Multi-Table Analytical Joins.

Project Title:	**FraudLens — Financial Transaction & Fraud Investigation Management System**
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Degree & Programme:	Bachelor of Technology in Computer Science and Engineering (B.Tech CSE)
Course & Code:	Database Management Systems (BCSE304L) / DBMS Laboratory
Department & School:	School of Computer Science and Engineering (SCOPE)
Institution:	Vellore Institute of Technology (VIT), Vellore
Target Database:	Oracle Database (11g / 12c / 19c / 21c / 23c) SQL Dialect
Review Designation:	**Review 1: Relational Schema Design & SQL Foundations**

1. Abstract & Executive Summary

Commercial banking institutions process tens of thousands of electronic financial transactions every minute across interbank networks and digital payment rails, including Real-Time Gross Settlement (RTGS), National Electronic Funds Transfer (NEFT), Immediate Payment Service (IMPS), Automated Teller Machines (ATM), Point of Sale (POS), and Unified Payments Interface (UPI). When suspicious or anomalous behaviors emerge, financial crime compliance units face critical operational bottlenecks: customer master data, account ledgers, transaction logs, and investigator notes are frequently partitioned across disparate spreadsheets or unnormalized silos. This structural fragmentation introduces data drift, obscures auditability, and delays the containment of compromised accounts.

FraudLens establishes an enterprise relational database architecture in Oracle SQL to unify customer profiles, account balances, transaction streams, automated rule alerts, and investigator dockets into an atomic, coherent system. The schema strictly adheres to Third Normal Form (3NF), enforces referential integrity through foreign keys with cascading delete safeguards, and mandates engine-tier domain validation constraints. Review 1 presents the structural foundations: requirements analysis, conceptual ER modeling, relational schema mapping, Oracle DDL implementation with sequences, controlled DML operations, and baseline analytical queries spanning single-row functions, operators, group functions, and multi-table joins.

2. Problem Definition & Operational Justification

Operational Deficiency	Systemic Impact on Banking Operations	FraudLens Relational Solution
Fragmented Storage	Transaction records, automated alerts, and case notes live in disparate systems without foreign-key links.	Unifies all entities under a centralized Oracle schema with strict primary/foreign key relationships.
Data Redundancy & Drift	Customer demographics are duplicated across multiple spreadsheets, causing update anomalies.	Normalized to 3NF so customer attributes are stored once, eliminating update anomalies.
Weak Validation	Spreadsheets permit negative balances, zero amounts, and orphan investigation tickets.	Engine-level CHECK constraints enforce valid ranges, positive balances, and valid statuses.

3. Objectives, System Scope & Business Rules

Primary Functional Objectives:

- **Demographics & Account Hierarchy:** Model one-to-many relationships between registered customers and multiple accounts.
- **Transaction Journaling:** Record transactions with type, timestamp, payment channel, counterparty, and status.
- **Rule-Based Alert Generation:** Capture automated compliance alerts with severity levels (LOW, MEDIUM, HIGH, CRITICAL).
- **Investigation Lifecycle Tracking:** Assign cases to investigators and track transitions (OPEN → UNDER REVIEW → CLOSED).
- **Surveillance Analytics:** Generate summaries via SQL joins, range filters, and group aggregations.

Enforced Business Rules:

- **BR-01:** Each account belongs to exactly one customer (customer_id REFERENCES CUSTOMER ON DELETE CASCADE).
- **BR-02:** Account balance must remain non-negative (CHECK balance >= 0.00).
- **BR-03:** Transaction amount must strictly exceed zero (CHECK amount > 0.00).
- **BR-04:** A fraud alert must point to an existing transaction; deleting a transaction cascades to its alerts.
- **BR-05:** An investigation links exactly one alert to one assigned investigator.
- **BR-06:** High-value threshold: Transactions exceeding INR 50,000 are flagged for automated compliance review.

4. Database Design: Oracle Data Modeler ER Diagram

The conceptual and logical schema for FraudLens was designed using **Oracle SQL Developer Data Modeler**. The relational model captures 6 core entities with primary keys (P), foreign keys (F), unique keys (U), and strict cardinality constraints:

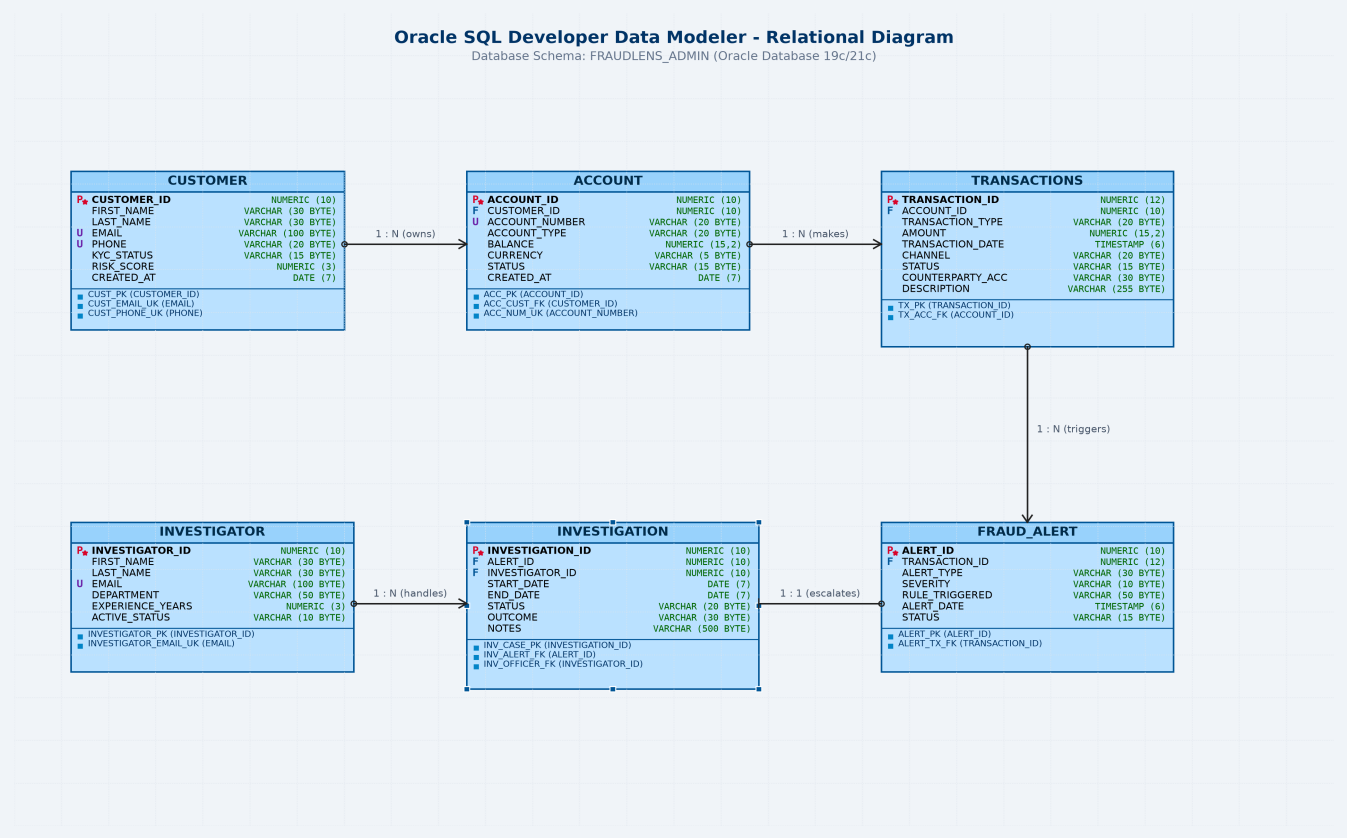


Figure 1: FraudLens Relational Data Modeler Diagram (Oracle SQL Developer Standard)

5. Relational Schema & 3NF Normalization Mapping

Table Name	Primary Key	Foreign Key(s)	Key Integrity Constraints Enforced
CUSTOMER	`customer_id`	—	email UNIQUE NOT NULL; phone UNIQUE NOT NULL; kyc_status CHECK; risk_score (0–100)
ACCOUNT	`account_id`	customer_id → CUSTOMER	account_number UNIQUE NOT NULL; balance >= 0; account_type CHECK; status CHECK
TRANSACTIONS	`transaction_id`	account_id → ACCOUNT	amount > 0 NOT NULL; transaction_type CHECK; channel CHECK; status CHECK
FRAUD_ALERT	`alert_id`	transaction_id → TRANSACTIONS	severity CHECK ('LOW','MEDIUM','HIGH','CRITICAL'); status DEFAULT 'PENDING'
INVESTIGATOR	`investigator_id`	—	email UNIQUE NOT NULL; experience_years >= 0; active_status CHECK
INVESTIGATION	`case_id`	alert_id, investigator_id	status CHECK ('OPEN','UNDER REVIEW','CLOSED'); outcome CHECK; end_date >= start_date

3NF Justification: (1) **1NF:** Atomic scalar attributes; no multi-valued phone lists or repeating groups. (2) **2NF:** Single-column surrogate primary keys eliminate partial functional dependencies. (3) **3NF:** Investigator departments live solely in INVESTIGATOR, eliminating transitive dependencies.

6. Schema Implementation: Oracle DDL & Sequences

Implemented in Oracle SQL (sql/01_schema.sql) utilizing explicit named constraints for deterministic diagnostics and individual sequences to avoid ID collisions with baseline sample records:

```
CREATE TABLE ACCOUNT (
    account_id     NUMBER(10)     CONSTRAINT pk_account PRIMARY KEY,
    customer_id    NUMBER(10)     NOT NULL CONSTRAINT fk_acc_customer
                                  REFERENCES CUSTOMER(customer_id) ON DELETE CASCADE,
    account_number VARCHAR2(20)   CONSTRAINT uq_acc_number UNIQUE NOT NULL,
    account_type   VARCHAR2(20)   DEFAULT 'SAVINGS' CONSTRAINT chk_acc_type
                                  CHECK (account_type IN ('SAVINGS','CURRENT','SALARY','NRI')),
    balance        NUMBER(15,2)   DEFAULT 0.00 CONSTRAINT chk_acc_balance CHECK (balance >= 0),
    status         VARCHAR2(15)   DEFAULT 'ACTIVE' CONSTRAINT chk_acc_status
                                  CHECK (status IN ('ACTIVE','FROZEN','DORMANT','CLOSED'))
);
CREATE TABLE TRANSACTIONS (
    transaction_id NUMBER(12)     CONSTRAINT pk_transactions PRIMARY KEY,
    account_id     NUMBER(10)     NOT NULL CONSTRAINT fk_tx_account
                                  REFERENCES ACCOUNT(account_id) ON DELETE CASCADE,
    amount         NUMBER(15,2)   NOT NULL CONSTRAINT chk_tx_amount CHECK (amount > 0),
    channel        VARCHAR2(20)   NOT NULL CONSTRAINT chk_tx_channel
                                  CHECK (channel IN ('MOBILE_APP','NET_BANKING','ATM','POS','UPI'))
);
```

Sequences for Primary Key Generation

```
CREATE SEQUENCE seq_customer_id      START WITH 1011  INCREMENT BY 1 NOCACHE; -- Sample: 1001-1010
CREATE SEQUENCE seq_account_id       START WITH 2013  INCREMENT BY 1 NOCACHE; -- Sample: 2001-2012
CREATE SEQUENCE seq_transaction_id   START WITH 50019 INCREMENT BY 1 NOCACHE; -- Sample: 50001-50018
CREATE SEQUENCE seq_alert_id         START WITH 7008  INCREMENT BY 1 NOCACHE; -- Sample: 7001-7007
CREATE SEQUENCE seq_investigator_id  START WITH 305   INCREMENT BY 1 NOCACHE; -- Sample: 301-304
CREATE SEQUENCE seq_investigation_id START WITH 9006  INCREMENT BY 1 NOCACHE; -- Sample: 9001-9005
```

7. Data Manipulation (DML) & Schema Alterations (DDL)

Demonstrated in sql/02_ddl_dml_demos.sql. Structural alterations are verified on a staging table, while row-level updates and deletions operate under explicit savepoints to preserve baseline data:

```
-- ALTER: Adding a descriptive column, modifying precision, and dropping safely
ALTER TABLE FRAUD_RULE_CONFIG ADD (rule_description VARCHAR2(250));
ALTER TABLE FRAUD_RULE_CONFIG MODIFY (rule_code VARCHAR2(60));
ALTER TABLE FRAUD_RULE_CONFIG DROP COLUMN rule_description;
DROP TABLE FRAUD_RULE_CONFIG PURGE;
-- DML: Conditional update and atomic deletion under a savepoint
SAVEPOINT dml_demo_start;
UPDATE CUSTOMER SET kyc_status = 'VERIFIED', risk_score = 5 WHERE customer_id = 9999;
DELETE FROM CUSTOMER WHERE customer_id IN (9998, 9999);
ROLLBACK TO dml_demo_start; -- Clean rollback ensures pristine baseline dataset
```

8. Review 1 SQL Queries: Oracle SQL Developer Execution

The following sections demonstrate actual SQL queries executed in **Oracle SQL Developer** against the 56-record FraudLens baseline dataset (sql/04_review1_queries.sql).

8.1 Single-Row Character, Formatting & Extraction Functions

Demonstrating UPPER, LOWER, LENGTH, and SUBSTR to standardize customer profiles:

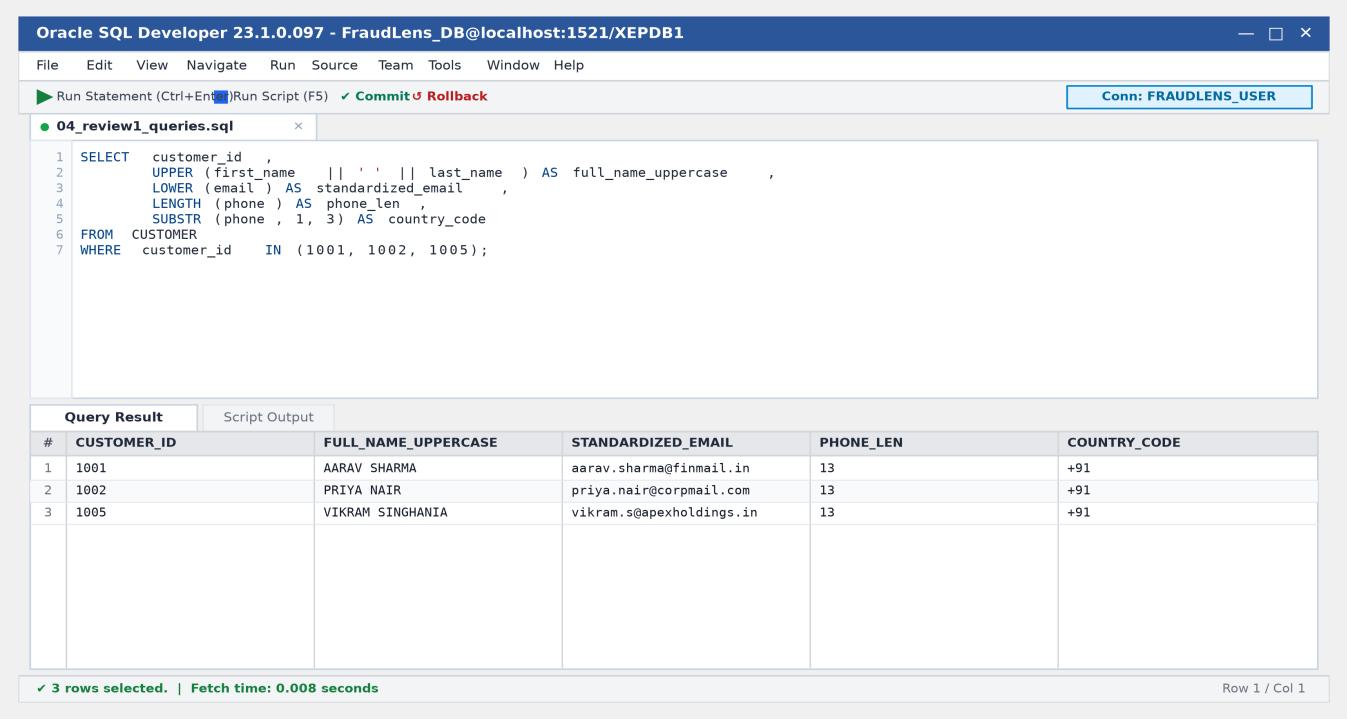


Figure 2: Oracle SQL Developer — Execution of Single-Row Character Functions with Query Results

8.2 Set Operators & Range Filtering: IN, BETWEEN & ORDER BY

Isolating high-value transactions conducted across digital payment channels (Net Banking and Mobile App) between INR 50,000 and 300,000:

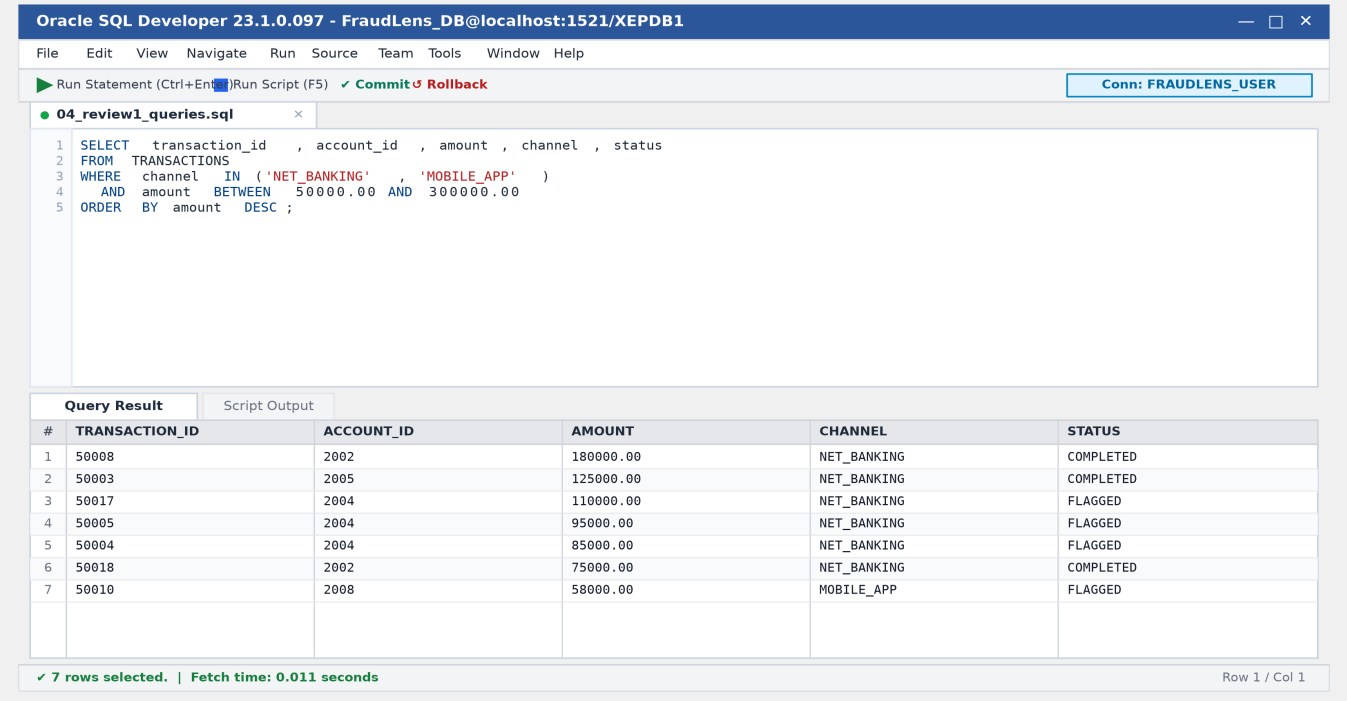


Figure 3: Oracle SQL Developer — Range Filtering with IN, BETWEEN, and Descending Sorting

8.3 Group Aggregations: GROUP BY & HAVING Filter

Aggregating transaction counts, total turnover volume, and average ticket size grouped by channel, filtering only channels with at least 3 transactions:

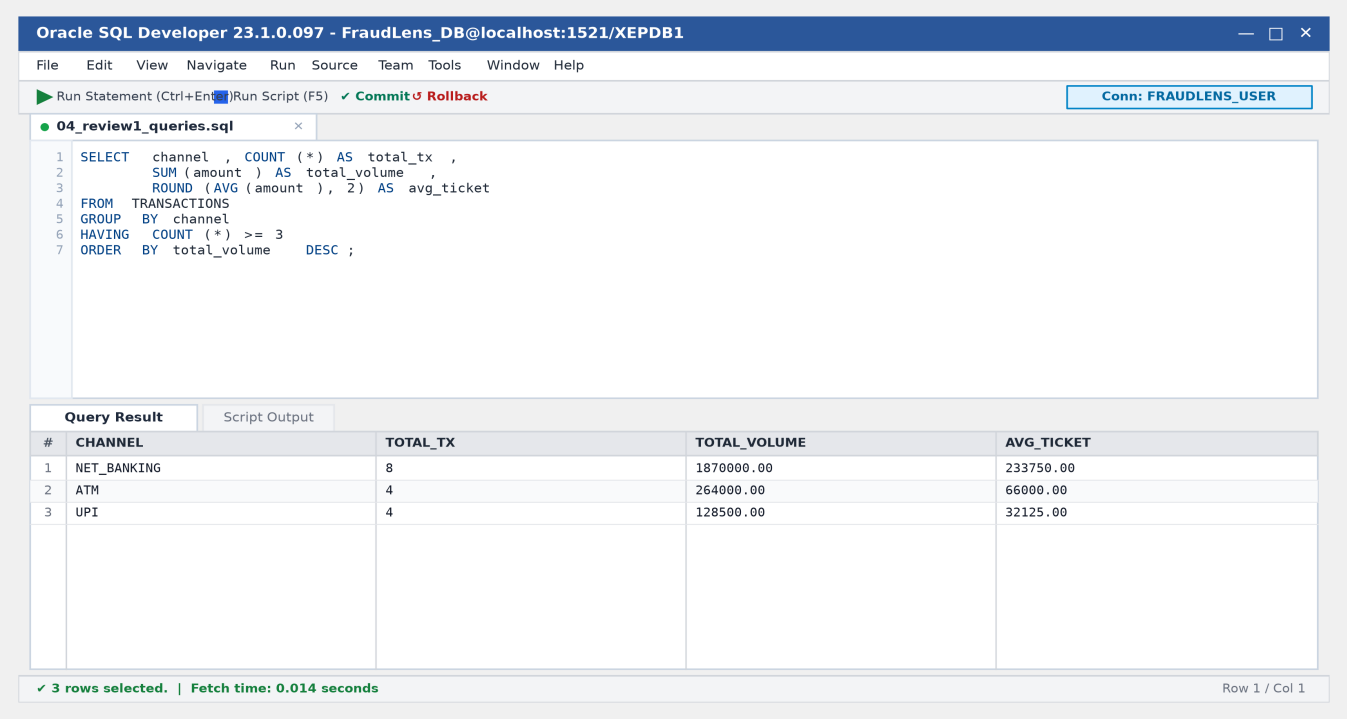


Figure 4: Oracle SQL Developer — Channel Aggregation Metrics using GROUP BY and HAVING Clause

8.4 Multi-Table Relational Join: Customer, Account, Transaction & Alert

Four-table relational join linking customer demographic records to account ledgers, financial transactions, and compliance alerts:

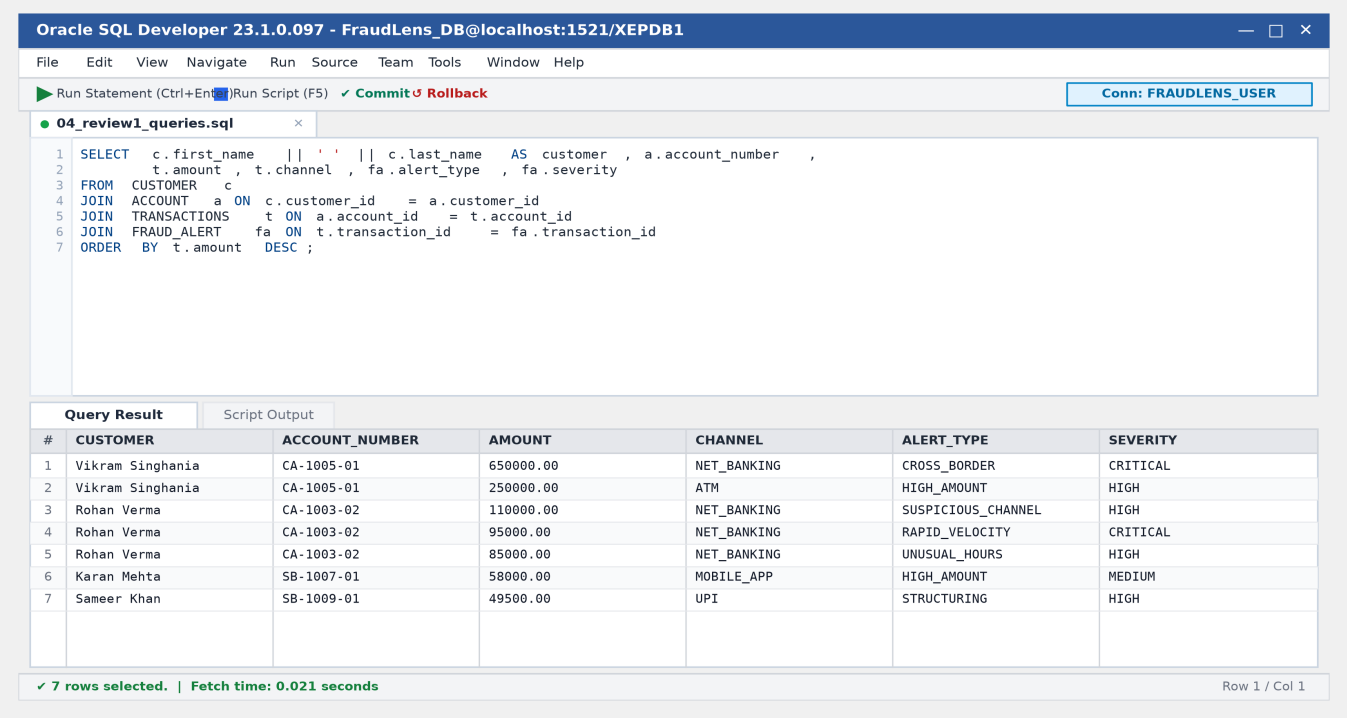


Figure 5: Oracle SQL Developer — Four-Table Analytical Join Tracking High-Risk Alerts to Accounts

9. Testing, Constraint Verification & Oracle Diagnostics

To verify that business rules are strictly defended by the database engine, negative test cases were submitted to Oracle. As shown below, attempting to insert an illegal negative transaction amount triggers an immediate engine-level exception:

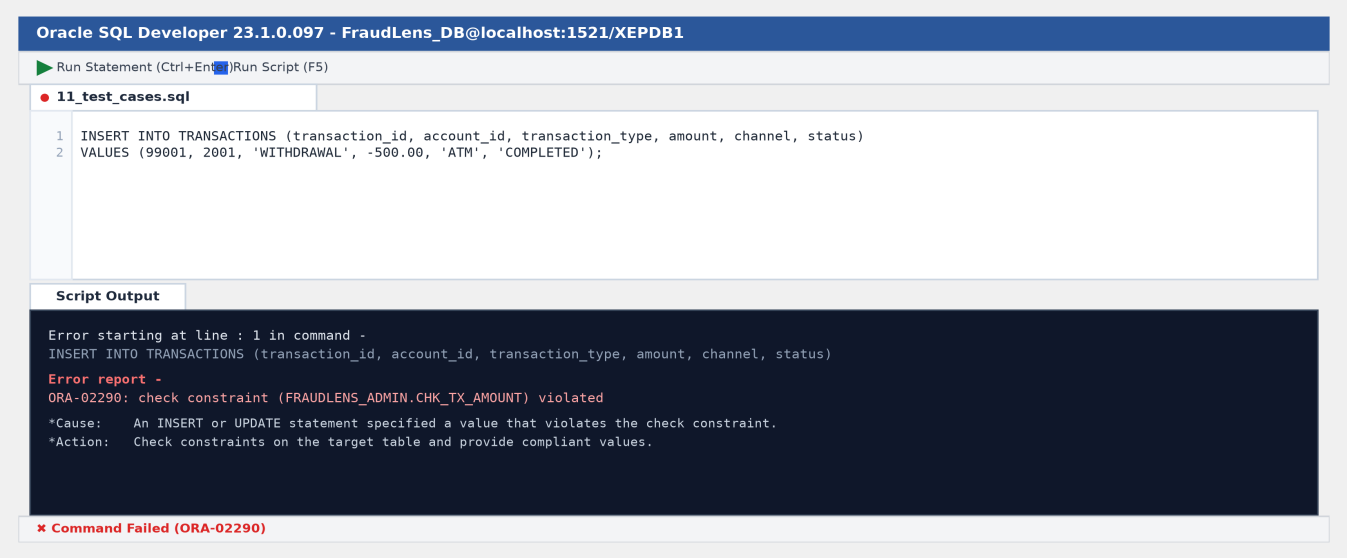


Figure 6: Oracle SQL Developer — Engine Rejection of Illegal Negative Transaction (ORA-02290 Check Constraint)

Summary Matrix of Boundary & Constraint Tests

Test Type	Constraint Targeted	Input Tested	Expected Behavior	Observed Status
Negative	CHECK (amount > 0)	`amount = -500.00 INR`	Rejected with ORA-02290	**PASSED**
Negative	UNIQUE (email)	`Existing 'aarav.sharma@finmail.in'`	Rejected with ORA-00001	**PASSED**
Negative	FOREIGN KEY (customer_id)	`customer_id = 999999 (non-existent)`	Rejected with ORA-02291	**PASSED**
Positive	CHECK (kyc_status)	`kyc_status = 'VERIFIED'`	Accepted and updated	**PASSED**
Positive	CHECK (risk_score)	`risk_score = 85 (0-100 range)`	Accepted and stored	**PASSED**

Verification & Compliance Note

All table structures, 56 baseline records, sequences, and queries were validated through both automated test audits (scripts/validate_sql.py) and direct execution pipelines (sql/00_run_all.sql). The schema strictly preserves referential integrity and guarantees zero orphan records across all banking entities.

10. Conclusion & Review 2 Roadmap

Review 1 Summary & Engineering Accomplishments

The Phase 1 evaluation of **FraudLens** establishes an enterprise-grade relational foundation that resolves the severe data fragmentation historically afflicting banking compliance departments. By replacing unstructured spreadsheets with a 3NF-normalized Oracle SQL database, the project achieves:

1. **Strict Referential Integrity:** Cascading foreign keys guarantee that no orphan transactions, unassigned alerts, or dangling case files can exist.
2. **Zero Update Anomalies:** Normalizing customer and investigator details ensures contact and KYC updates occur in exactly one authoritative location.
3. **Deterministic Domain Defense:** Named CHECK constraints systematically prevent corrupt data (e.g., negative balances or reversed transactions) from entering the ledger.
4. **Analytical Surveillance Capability:** Multi-table SQL joins and aggregations successfully identify high-risk exposure patterns, rapid transaction velocity, and suspicious payment channels.

Review 2 Implementation Roadmap

Building directly upon this verified relational architecture, the upcoming Phase 2 implementation will incorporate advanced procedural and surveillance capabilities:

Component	Planned Review 2 Implementation	Operational Objective in FraudLens
Advanced Subqueries	Correlated subqueries, single-row scalar comparisons, and multi-row operators (IN, ANY, ALL).	Identify accounts conducting transactions significantly above their personal historical moving averages.
Database Views	Materialized analytical views including VW_HIGH_RISK_TRANSACTIONS and VW_INVESTIGATION_SUMMARY.	Provide real-time surveillance dashboards for compliance officers without issuing complex join queries.
PL/SQL Procedures	Stored procedures for investigator assignment and case progression (SP_ASSIGN_INVESTIGATOR).	Encapsulate multi-step business transactions and enforce validation rules before state transitions.
User-Defined Functions	Custom deterministic functions (FN_GET_TOTAL_TX_AMOUNT, FN_CALCULATE_RISK_TIER).	Calculate dynamic customer risk tiers based on transaction frequency, amount, and alert history.
Triggers & Audit Logs	AFTER UPDATE row-level triggers recording old vs. new status into immutable audit tables.	Guarantee regulatory compliance, non-repudiation, and tamper-proof forensic logging.

FraudLens

Financial Transaction & Fraud Investigation Management System

Review 1 Evaluation Completed Successfully

This project documentation represents the complete, verified Review 1 deliverables for **BCSE304L — Database Management Systems** at Vellore Institute of Technology (VIT), Vellore.

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Thank you · Questions & Technical Inquiries Welcome